

MATH 324 Winter 2025 - Tutorial 2

Section 003 (Thursday)

January 23, 2025

1 Bias and Mean Squared Error

Definition: We say that an estimator $\hat{\theta}$ is *unbiased* for the parameter θ if $\mathbb{E}[\hat{\theta}] = \theta$.

Definition: The *mean-squared error* of an estimator $\hat{\theta}$ is

$$\text{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2].$$

Question 1 - Binomial bias

(Exercise 8.17) If Y has a binomial distribution with parameters n and p , then $\hat{p}_1 = Y/n$ is an unbiased estimator of p . Another estimator of p is $\hat{p}_2 = (Y + 1)/(n + 2)$.

- (a) Derive the bias of $\hat{p}_2$.
- (b) Derive $\text{MSE}(\hat{p}_1)$ and $\text{MSE}(\hat{p}_2)$.
- (c) For what values of p is $\text{MSE}(\hat{p}_1) < \text{MSE}(\hat{p}_2)$?

Solution

- (a) $B(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$.

$$\begin{aligned}\mathbb{E}[\hat{p}_2] &= \mathbb{E}\left[\frac{Y+1}{n+2}\right] = \frac{1}{n+2}(\mathbb{E}[Y] + 1) = \frac{np+1}{n+2} \\ B(\hat{p}_2) &= \frac{np+1}{n+2} - p = \frac{1-2p}{n+2}\end{aligned}$$

- (b) $\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) - (B(\hat{\theta}))^2$

As $\hat{p}_1$ is unbiased, $\text{MSE}(\hat{p}_1) = \text{Var}(\hat{p}_1) = \frac{p(1-p)}{n}$.

To find $\text{MSE}(\hat{p}_2)$, we first find

$$\text{Var}(\hat{p}_2) = \text{Var}\left(\frac{Y+1}{n+2}\right) = \frac{1}{(n+2)^2} \text{Var}(Y+1) = \frac{1}{(n+2)^2} \text{Var}(Y) = \frac{np(1-p)}{(n+2)^2}$$

Then,

$$\text{MSE}(\hat{p}_2) = \frac{np(1-p)}{(n+2)^2} + \left(\frac{1-2p}{n+2}\right)^2 = \frac{np(1-p) + (1-2p)^2}{(n+2)^2}$$

(c) We search for values of p where

$$\frac{p(1-p)}{n} < \frac{np(1-p) + (1-2p)^2}{(n+2)^2}$$

By expanding the products and grouping the p terms, we can find that

$$(8+4n)p^2 - (8n+5)p + n < 0$$

which is positive quadratic in p . This means that the values between the roots will be negative. We can solve for p using quadratic formula:

$$p = \frac{8n+4 \pm \sqrt{(8n+4)^2 - 4(8n+4)n}}{2(8n+4)} = \frac{1}{2} \pm \sqrt{\frac{n+1}{8n+4}}$$

This is approximately $\frac{1}{2} \pm \sqrt{\frac{1}{8}}$.

Question 2 - Estimating a mean

(Exercise 8.22) An increase in the rate of consumer savings frequently is tied to a lack of confidence in the economy and is said to be an indicator of a recessionary tendency in the economy. A random sampling of $n = 200$ savings accounts in a local community showed the mean increase in savings account values to be 7.2% over the past 12 months, with standard deviation 5.6%. Estimate the mean percentage increase in savings account values over the past 12 months for depositors in the community. Place a bound on your error of estimation.

Solution

We have $\bar{y} = 0.072$ and $\hat{\sigma} = 0.056$. Thus,

$$\sigma_{\bar{y}} = \frac{\hat{\sigma}}{\sqrt{n}} = \frac{0.056}{\sqrt{200}}.$$

Our estimated mean percentage $\hat{\mu} = \bar{y} = 0.072$. Our 2 standard error bound is

$$\hat{\mu} \pm 2\sigma_{\bar{y}} = 0.072 \pm \frac{2(0.056)}{\sqrt{200}} = 0.072 \pm 0.0079.$$

Question 3 - Estimating an exponential mean parameter

(Exercise 8.36-8.37) If $Y_1, Y_2, \dots, Y_n$ denote a random sample from an exponential distribution with mean θ , then $\mathbb{E}[Y_i] = \theta$ and $\text{Var}(Y_i) = \theta^2$.

- Suggest an unbiased estimator for θ and provide an estimate for the standard error of your estimator. (*Hint:* Note that θ is the mean!)
- An engineer observes $n = 10$ independent length-of-life measurements on a type of electronic component. The average of these 10 measurements is 1020 hours. If these lengths of life come from an exponential distribution with mean θ , estimate θ and place a 2-standard-error bound on the error of estimation.

Solution

- (a) $\mathbb{E}[\bar{Y}] = \mathbb{E}[Y_i] = \theta$, so $\bar{Y}$ is unbiased for θ . Meanwhile, $\text{Var}(\bar{Y}) = \frac{1}{n} \text{Var}(Y_i) = \frac{\theta^2}{n}$. Thus, our standard error is

$$\sigma_{\hat{\theta}} = \sqrt{\text{Var}(\bar{Y})} = \frac{\theta}{\sqrt{n}}.$$

- (b) From the data, $\hat{\theta} = \bar{Y} = 1020$ and $\sigma_{\hat{\theta}} = \frac{1020}{\sqrt{n}}$. Our estimate is thus

$$1020 \pm \frac{2(1020)}{\sqrt{n}} = 1020 \pm 645.1,$$

which represents the mean length of life for the component in hours.

2 Error Bounds and Confidence Intervals

2.1 Confidence interval

A $100(1 - \alpha)\%$ confidence interval is a random interval such that the probability that the true parameter θ falls within this interval is $1 - \alpha$. In a two-sided confidence interval with upper and lower limits of $[\hat{\theta}_L, \hat{\theta}_U]$, this means

$$\mathbb{P}(\hat{\theta}_L \leq \theta \leq \hat{\theta}_U) = 1 - \alpha.$$

The confidence interval is constructed using the sample data; the $1 - \alpha$ reflects the amount of time the interval is expected to capture the true parameter.

There are also one-sided confidence intervals. The lower confidence interval is $[\hat{\theta}_L, \infty)$ such that

$$\mathbb{P}(\hat{\theta}_L \leq \theta) = 1 - \alpha,$$

and the upper confidence interval is $(-\infty, \hat{\theta}_U]$ such that

$$\mathbb{P}(\theta \leq \hat{\theta}_U) = 1 - \alpha.$$

Definition: A *pivotal quantity* for a parameter θ is a function of θ and the sample data such that

- (1) The only unknown quantity is θ , and
- (2) Its probability distribution does not depend on θ .

2.2 Central limit theorem

Recall that for large samples,

$$Z = \frac{\hat{\theta} - \mathbb{E}[\hat{\theta}]}{\sigma_{\hat{\theta}}} \approx N(0, 1)$$

For an unbiased estimator, then $\mathbb{E}[\hat{\theta}] = \theta$.

Thus Z is a pivotal quantity and we can construct a two-sided confidence interval as

$$\hat{\theta} \pm z_{\alpha/2} \sigma_{\hat{\theta}}.$$

The “ $z_{\alpha/2} \sigma_{\hat{\theta}}$ ” term is the sampling error, or margin of error, which forms the error bounds around our point estimate. Generally, increasing sample size helps reduce this M.O.E.

The lower and upper one-sided intervals would use either $\hat{\theta} + z_{\alpha} \sigma_{\hat{\theta}}$ or $\hat{\theta} - z_{\alpha} \sigma_{\hat{\theta}}$, respectively.

2.3 Small samples and the t-distribution

We consider the case of estimating the mean. If the population variance is unknown and the sample size is too low to use central limit theorem, then

$$T = \frac{\bar{X}_n - \mu}{S/\sqrt{n}} \sim t_{n-1}$$

a t-distribution with $n - 1$ degrees of freedom. We use T instead of Z , but proceed similarly.

Question 4 - Estimating a difference of proportions

(Exercise 8.28) In a study of the relationship between birth order and college success, an investigator found that 126 in a sample of 180 college graduates were firstborn or only children; in a sample of 100 nongraduates of comparable age and socioeconomic background, the number of firstborn or only children was 54. Estimate the difference in the proportions of firstborn or only children for the two populations from which these samples were drawn. Give a bound for the error of estimation.

Solution

We are looking for $\widehat{p_1 - p_2}$. Firstly, $\hat{p}_1 = \frac{126}{180} = 0.7$ and $\hat{p}_2 = \frac{54}{100} = 0.54$. Thus $\widehat{p_1 - p_2} = \hat{p}_1 - \hat{p}_2 = 0.16$.

We are going to use a 2σ bound for the error of estimation. From Table 8.1 of the textbook,

$$\sigma_{\widehat{p_1 - p_2}} = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}} = \sqrt{\frac{0.7(0.3)}{180} + \frac{0.54(0.46)}{100}} = .0604$$

Therefore, our error of estimation is about 0.12.

Question 5 - Pivotal quantities

(Exercise 8.47) Assume that $Y_1, Y_2, \dots, Y_n$ is a sample of size n from an exponential distribution with mean θ .

- Use the method of moment-generating functions to show that $2 \sum_{i=1}^n Y_i / \theta$ is a pivotal quantity and has a χ^2 distribution with $2n$ df.
- Use the pivotal quantity $2 \sum_{i=1}^n Y_i / \theta$ to derive a 95% confidence interval for θ .
- If a sample of size $n = 7$ yields $\bar{y} = 4.77$, use the result from part (b) to give a 95% confidence interval for θ .

Solution

- (a) For all i , the mgf for Y_i is $M_Y(t) = (1 - \theta t)^{-1}, t < 1/\theta$.

Let $U = 2 \sum_{i=1}^n Y_i/\theta$. We find its mgf:

$$\begin{aligned} M_U(t) &= \mathbb{E}[e^{Ut}] = \mathbb{E}[e^{(2 \sum Y_i/\theta)t}] \\ &= \mathbb{E}[e^{2t/\theta \sum Y_i}] \\ &= \mathbb{E}\left[\prod_{i=1}^n e^{(2t/\theta)Y_i}\right] \\ &= \prod_{i=1}^n \mathbb{E}[e^{(2t/\theta)Y_i}] \quad \text{by independence} \\ &= [m_Y(2t/\theta)]^n \\ &= (1 - 2t)^{-n} \end{aligned}$$

This is the mgf for the chi-square with $2n$ degrees of freedom. Thus, the distribution of U does not depend on θ and shows that U is a pivotal quantity.

- (b) Following the textbook notation, let $\chi_{\alpha,k}^2$ be the value such that if X is a χ^2 random variable with k degrees of freedom, then

$$\mathbb{P}(X \geq \chi_{\alpha,k}^2) = \alpha.$$

Then we can have that

$$\mathbb{P}\left(\chi_{0.975,2n}^2 \leq 2 \sum_{i=1}^n Y_i/\theta \leq \chi_{0.025,2n}^2\right) = 0.95$$

and $\left(\frac{2 \sum_{i=1}^n Y_i}{\chi_{0.975,2n}^2}, \frac{2 \sum_{i=1}^n Y_i}{\chi_{0.025,2n}^2}\right)$ forms our 95% confidence interval for θ .

- (c) Looking at the table, $\chi_{0.975,14}^2 = 5.629$ and $\chi_{0.975,14}^2 = 26.119$. Thus, our confidence interval is

$$\left(\frac{2(7)(4.77)}{5.629}, \frac{2(7)(4.77)}{26.119}\right) = (2.557, 11.864).$$

Question 6 - Confidence interval for large sample

(Exercise 8.65) For a comparison of the rates of defectives produced by two assembly lines, independent random samples of 100 items were selected from each line. Line A yielded 18 defectives in the sample, and line B yielded 12 defectives.

- (a) Find a 98% confidence interval for the true difference in proportions of defectives for the two lines.
(b) Is there evidence here to suggest that one line produces a higher proportion of defectives than the other?

Solution

- (a) We need $z_{\alpha/2} = z_{0.01} = 2.326$. In the case of difference of proportions, recall that $\sigma_p^2 = \frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}$. In this case, $n_1 = n_2 = 100$. Thus, our 98% confidence interval is

$$0.18 - 0.12 \pm 2.326 \sqrt{\frac{0.18(1-0.18) + 0.12(1-0.12)}{100}} = (-0.057, 0.177).$$

- (b) Our confidence interval contains 0, so it is likely the two assembly lines produce the same proportion of defectives.

Question 7 - Confidence interval for small sample

(Exercise 8.94) Suppose that we obtain independent samples of sizes n_1 and n_2 from two normal populations with equal variances. Use an appropriate pivotal quantity to derive a $100(1 - \alpha)\%$ upper confidence bound for $\mu_1 - \mu_2$.

Solution

From section 8.8, we know that the pivotal quantity for difference in means with a small sample is

$$T = \frac{\bar{Y}_1 - \bar{Y}_2 - (\mu_1 - \mu_2)}{S_p \sqrt{1/n_1 + 1/n_2}}$$

which has a t-distribution w/ $n_1 + n_2 - 2$ degrees of freedom. We are looking for the upper confidence bound, i.e. t_α such that $\mathbb{P}(T > t_\alpha) = \alpha$.

Following the usual confidence interval approach, our $100(1 - \alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is

$$\bar{Y}_1 - \bar{Y}_2 \pm t_\alpha S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}.$$